Department of Computer Engineering COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Implementation of Differentiator and Accumulator Systems for Discrete-Time Signals.
Lab Number	**04**
Submitted by:	
Student Names	**Registration #**
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Rubrics		**Marks**		
		1	**2**	**3**
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

Lab-04

Digital Signal Processing - Discrete-Time Signal Operations

Objective:

To implement and analyze two fundamental discrete-time systems:

- **Differentiator**: Computes the first difference of a signal.
- **Accumulator**: Computes the running sum of a signal.

Understand the behavior and characteristics of these systems through MATLAB implementation and visualization.

Lab Task:

1. Consider the discrete-time signal:

$$x(n) = \begin{cases} |n|, & -3 \le n \le 3 \\ 0, & \text{otherwise} \end{cases}$$

(a) Implement a differentiator defined as:

$$y[n] = x[n] - x[n-1], \quad \text{with } x[-1] = 0$$

(b) Implement an accumulator defined as:

$$z[n] = \sum_{k=0}^{n} x[k]$$

2. Plot the original signal x[n], the differentiated signal y[n], and the accumulated signal z[n] using stem plots.

3. Comment on the results:

o How does the differentiator respond to changes in the signal?

o How does the accumulator modify the signal?

Code:

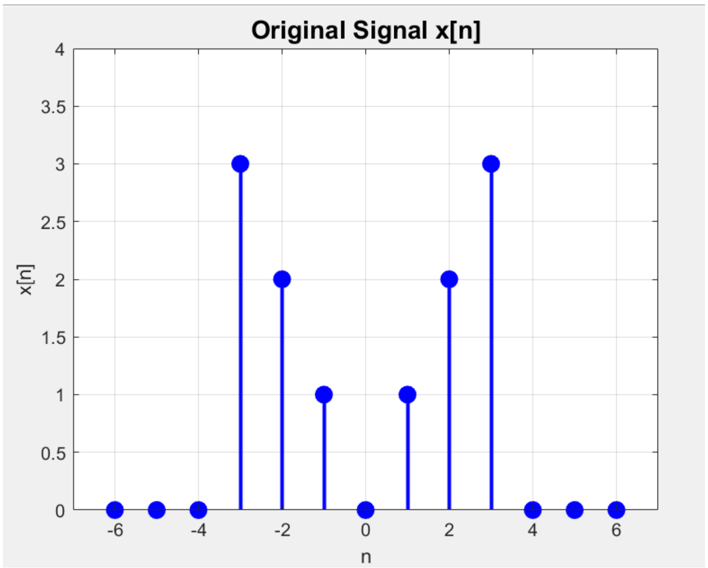
```
clc; clear; close all;
% 1. Define the discrete-time signal x[n]
n = -6:6;
x = zeros(size(n));
% Assign values for -3 ? n ? 3
for i = 1:length(n)
   if n(i) >= -3 && n(i) <= 3
   x(i) = abs(n(i));  % x[n] = |n|
   else
    x(i) = 0;         % otherwise 0
   end
```

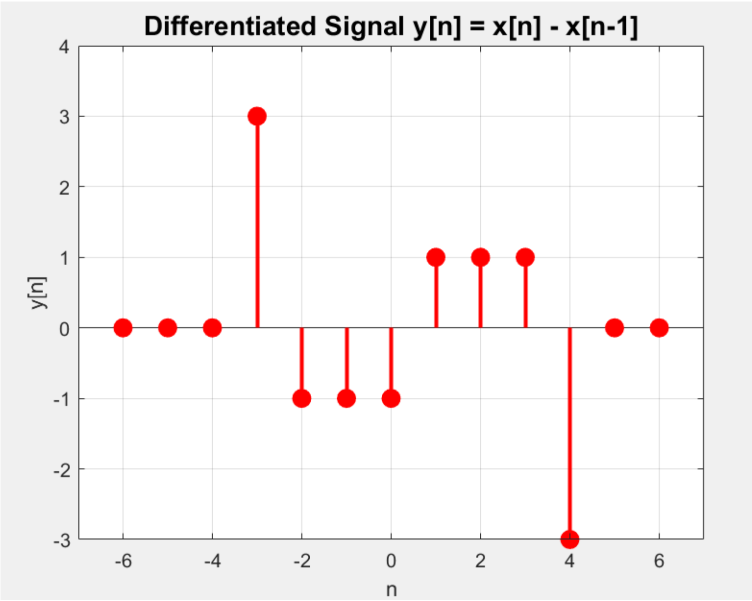
```
end
% 2. Differentiator: y[n] = x[n] - x[n-1], with x[-7] = 0
y = zeros(size(x));
for i = 1:length(n)
   if n(i)-1 >= -6
   prev_idx = find(n == n(i)-1, 1);
   y(i) = x(i) - x(prev_idx);
   else
   y(i) = x(i);  % for n = -6, assume x[-7] = 0
   end
end
% 3. Accumulator: z[n] = ? x[k], from k=0 to n (causal)
z = zeros(size(x));
for i = 1:length(n)
   if n(i) >= 0
   total = 0;
   for k = 0:n(i)
   k_idx = find(n == k, 1);
   total = total + x(k_idx);
   end
   z(i) = total;
   else
   z(i) = 0; % for n < 0
   end
end
% 4. Display results
disp('---------------------------------------------');
disp('   n       x[n]       y[n]       z[n]');
disp('---------------------------------------------');
disp([n' x' y' z']);
disp('---------------------------------------------');

% --- Original Signal ---
figure('Name','Original Signal','NumberTitle','off');
stem(n, x, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'blue');
title('Original Signal x[n]', 'FontSize', 14, 'FontWeight', 'bold');
xlabel('n'); ylabel('x[n]');
grid on; ylim([0 4]); xlim([-7 7]);

% --- Differentiated Signal ---
figure('Name','Differentiated Signal','NumberTitle','off');
stem(n, y, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'red');
title('Differentiated Signal y[n] = x[n] - x[n-1]', 'FontSize', 14, 'FontWeight', 'bold');
xlabel('n'); ylabel('y[n]');
grid on; ylim([-3 4]); xlim([-7 7]);

% --- Accumulated Signal ---
figure('Name','Accumulated Signal','NumberTitle','off');
```

```
stem(n, z, 'filled', 'LineWidth', 2, 'MarkerSize', 8, 'Color', 'green');
title('Accumulated Signal z[n] = ? x[k] (k=0?n)', 'FontSize', 14, 'FontWeight', 'bold');
xlabel('n'); ylabel('z[n]');
grid on; ylim([0 7]); xlim([-7 7]);
```

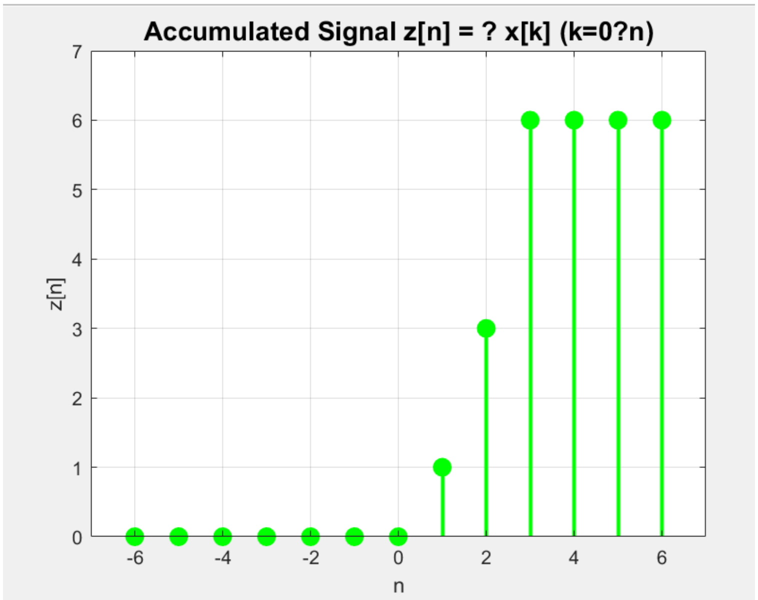
Output:

Original Signal:





Accumulated Signal:


Accumulated Signal z[n] = ? x[k] (k=0?n)
z[n]
n
7
6
5
4
3
2
1
0
-6
-4
-2
0
2
4
6